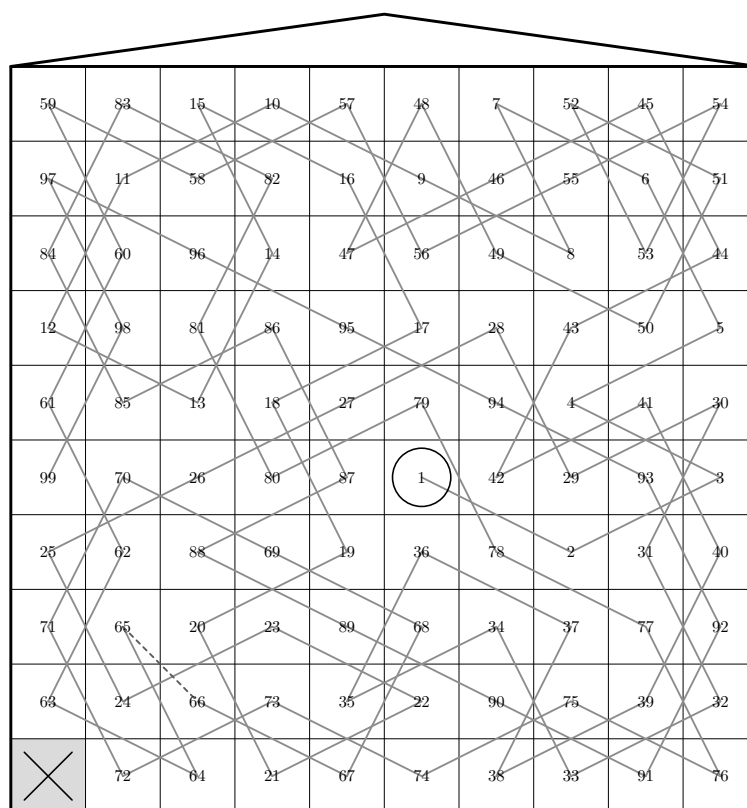


1. The knight's tour. Georges Perec's novel "*Life A User's Manual (La Vie mode d'emploi, 1978)*" is set in an apartment building at 11 rue Simon-Crubbellier in Paris. Perec cut the building's facade away like a doll's house, imagining a $10 \times 10 = 100$ -cell grid from the cellars to the attic. Each chapter of the novel dwells in one of those 100 cells and tells the story unfolding in that room.

The question is the order in which the narrative moves from room to room. Perec fixed it by the *knight's move* of chess: he chose a knight's tour of the 10×10 board—a path that steps by knight's moves through all 100 cells, never landing on one twice—and let the chapters travel in that order. The first chapter begins at the central landing (6, 6).

Yet the novel has 99 chapters, not 100. Partway through the tour Perec deliberately skipped a cell: the cellar at the bottom-left corner (1, 10). Borrowing a word from Lucretius, he called this deliberate flaw the *clinamen* (the slight swerve of an atom from its ordained path). Because of it, the move from the 65th chapter to the 66th is not a knight's move but an illegal diagonal step of one cell—the single blemish on an otherwise perfect tour.

Here is a picture. The facade is cut away like a doll's house so that all 100 rooms lie open at once; each cell is stamped with its chapter number, and consecutive chapters are joined by a line. The first chapter is the ringed 1 at the center, and the bottom-left corner—the unvisited clinamen—is left empty with a cross. The dashed line from 65 to 66 is that one illegal, non-knight move.



Perec had a second constraint. He gathered 42 lists of ten items into twenty-one pairs and, by an order-10 Graeco-Latin square, assigned to each chapter a combination of items. If the knight's tour decides *where to write*, this square decides *what to write*. This starred section builds and verifies the *knight's tour*; the square that distributes the material is treated in a later starred section.

2. Here is what the program does. Using GB.BASIC's *Board* it builds the 10×10 knight board, lays Perec's actual chapter order on it, and verifies—by asking the board's own arcs—that the order really is a walk of knight's moves, with the clinamen as its one exception. Finally it prints a grid of chapter numbers and a diagnosis.

The chapter-order data is transcribed from the *sqs* array in `scripts/knights-tour.js`, published by Thomas Guest at wordaligned.org/knights-tour. Whether the transcription is faithful the program checks for itself, on the board.

```
package main
import (
    "bufio"
    "fmt"
    "log"
    "os"

    "github.com/sjnam/go-sgb/gbbasic"
)
<Types and data 3>
func main() {
    <Build the knight board 4>
    <Extract the board's adjacency 5>
    out := bufio.NewWriter(os.Stdout)
    defer out.Flush()
    <Verify and print the tour 7>
    <Generate and print a complete tour 15>
    <Verify the square and print the assignment 26>
}
```

3. A cell of the board is a **cell**. Here x is the column (1 at the left, 10 at the right) and y the floor (1 at the top, 10 at the bottom). `tour[k]` is the cell the $(k + 1)$ th chapter sits in.

<Types and data 3> $\equiv$

```
type cell struct { x, y int }
var tour = []cell{
    {6, 6}, {8, 7}, {10, 6}, {8, 5}, {10, 4}, {9, 2}, {7, 1}, {8, 3}, {6, 2}, {4, 1},
    {2, 2}, {1, 4}, {3, 5}, {4, 3}, {3, 1}, {5, 2}, {6, 4}, {4, 5}, {5, 7}, {3, 8},
    {4, 10}, {6, 9}, {4, 8}, {2, 9}, {1, 7}, {3, 6}, {5, 5}, {7, 4}, {8, 6}, {10, 5},
    {9, 7}, {10, 9}, {8, 10}, {7, 8}, {5, 9}, {6, 7}, {8, 8}, {7, 10}, {9, 9}, {10, 7},
    {9, 5}, {7, 6}, {8, 4}, {10, 3}, {9, 1}, {7, 2}, {5, 3}, {6, 1}, {7, 3}, {9, 4},
    {10, 2}, {8, 1}, {9, 3}, {10, 1}, {8, 2}, {6, 3}, {5, 1}, {3, 2}, {1, 1}, {2, 3},
    {1, 5}, {2, 7}, {1, 9}, {3, 10}, {2, 8}, {3, 9}, {5, 10}, {6, 8}, {4, 7}, {2, 6},
    {1, 8}, {2, 10}, {4, 9}, {6, 10}, {8, 9}, {10, 10}, {9, 8}, {7, 7}, {6, 5}, {4, 6},
    {3, 4}, {4, 2}, {2, 1}, {1, 3}, {2, 5}, {4, 4}, {5, 6}, {3, 7}, {5, 8}, {7, 9},
    {9, 10}, {10, 8}, {9, 6}, {7, 5}, {5, 4}, {3, 3}, {1, 2}, {2, 4}, {1, 6},
}
```

This code is also defined in sections 6, 12, 24.

This code is used in section 2.

4. *Board* builds a board graph from the moves of a generalized chess piece. Its first four arguments are the board's dimensions; a zero dimension is unused, so 10, 10, 0, 0 is a two-dimensional 10 × 10 board. The fifth argument *piece* = 5 is the knight—a knight's move is exactly the one whose Euclidean distance between two cells is $\sqrt{5}$. The last two arguments are wrapping and directedness, both unused here.

⟨Build the knight board 4⟩ ≡

```
g, err := gbbasic.Board(10, 10, 0, 0, 5, 0, false)
if err != Λ {
    log.Fatalf("couldn't build the knight board: %v", err)
}
```

This code is used in section 2.

5. The vertices *Board* makes are named by coordinate "row.col", so the cell at floor *y* and column *x* is "y-1.x-1" (both counted from 0). Verification needs not the vertices themselves but only whether two cells are neighbors on the board, so we sweep all of the board's arcs into a single set of name pairs. Then one lookup tells whether two cells are joined by a knight's move.

⟨Extract the board's adjacency 5⟩ ≡

```
adj := make(map[[2]string]bool)
for v := range g.AllVertices() {
    for a := range v.AllArcs() {
        adj[[2]string{v.Name, a.Tip.Name}] = true
    }
}
```

This code is used in section 2.

6. This *name* is the bridge from a coordinate to a vertex name: floor *y* first, column *x* second, each decremented to count from 0.

⟨Types and data 3⟩ +≡

```
func name(c cell) string { return fmt.Sprintf("%d.%d", c.y-1, c.x-1) }
```

7. Now the heart of it. First we sweep Perec's chapter order as a walk on the board, collecting the links that are not neighbors and the cells stepped on twice. Then we pick out the cell, among the 100, that is never stepped on. If Perec's tour is right, the only non-adjacent link is the clinamen, and the only unvisited cell is the cellar corner.

⟨Verify and print the tour 7⟩ ≡

```
fmt.Fprint(out, "PEREC: the knight's tour of Life A User's Manual\n\n")
fmt.Fprintf(out, "The board's official name is\n\n%s\n", g.ID)
fmt.Fprintf(out, "with %d vertices and %d arcs.\n\n", g.N, g.M)
⟨Find non-adjacent links and repeated cells 8⟩
⟨Find the unvisited cell 9⟩
⟨Print the chapter-number grid 10⟩
⟨Print the verification result 11⟩
```

This code is used in section 2.

8. For each consecutive pair of chapters we ask *adj* whether their cells are neighbors on the board. If not, we record the link in *breaks*. We also record in *chapterAt* the first chapter to step on each cell, so a cell stepped on twice shows up at once.

⟨Find non-adjacent links and repeated cells 8⟩ ≡

```
chapterAt := make(map[string]int)
var breaks [[2]int // (earlier chapter, later chapter) of a non-adjacent link
var repeats int
for k, c := range tour {
    if _, seen := chapterAt[name(c)]; !seen {
        repeats++
    }
    chapterAt[name(c)] = k + 1
    if k > 0 ∧ ¬adj[[2]string{name(tour[k-1]), name(c)}] {
        breaks = append(breaks, [2]int{k, k+1})
    }
}
```

This code is used in section 7.

9. We walk all 100 cells of the board and gather those not in *chapterAt*, forming the coordinates by *Board*'s naming rule from floor *y* and column *x*.

⟨Find the unvisited cell 9⟩ ≡

```
var missing []cell
for y := 1; y ≤ 10; y++ {
    for x := 1; x ≤ 10; x++ {
        if _, seen := chapterAt[name(cell{x, y})]; !seen {
            missing = append(missing, cell{x, y})
        }
    }
}
```

This code is used in section 7.

10. The chapter-number grid is a copy of the novel's facade. From the top-left (1,1) to the bottom-right (10,10), each cell is stamped with the chapter that treats that room. The unvisited cellar corner is left blank with three dots.

⟨Print the chapter-number grid 10⟩ ≡

```
fmt.Fprint(out, "Chapter-number grid (top-left is (1,1), top row is the attic):\n\n")
for y := 1; y ≤ 10; y++ {
    fmt.Fprint(out, " ")
    for x := 1; x ≤ 10; x++ {
        if ch, seen := chapterAt[name(cell{x, y})]; seen {
            fmt.Fprintf(out, "%3d", ch)
        } else {
            fmt.Fprint(out, ". . .")
        }
    }
    fmt.Fprint(out, "\n")
}
fmt.Fprint(out, "\n")
```

This code is used in section 7.

11. Finally we put the verification into words a reader can follow. For each non-adjacent link we tell between which chapters, and to which cell, it strayed, and whether the stray is a single diagonal step. Perc's clinamen is exactly that one move from the 65th chapter to the 66th.

⟨Print the verification result 11⟩ ≡

```
fmt.Fprintf(out, "%d chapters: %d, repeated cells: %d\n", len(tour), repeats)
for _, m := range missing {
    fmt.Fprintf(out, "%d unvisited cell: (%d,%d) <- clinamen\n", m.x, m.y)
}
for _, b := range breaks {
    p, q := tour[b[0]-1], tour[b[1]-1]
    dx, dy := abs(p.x - q.x), abs(p.y - q.y)
    fmt.Fprintf(out,
        "%d non-knight move: ch.%d (%d,%d) -> ch.%d (%d,%d), offset (%d,%d)\n",
        b[0], p.x, p.y, b[1], q.x, q.y, dx, dy)
}
```

This code is used in section 7.

12. A small hand for the absolute value of a coordinate difference.

⟨Types and data 3⟩ +≡

```
func abs(n int) int {
    if n < 0 {
        return -n
    }
    return n
}
```

13. Run the program and it prints this for the tour: the board's official name and its vertex and arc counts, the chapter-number grid modeled on the novel's facade, and then the verification result.

PEREC: the knight's tour of Life A User's Manual

The board's official name is
board(10,10,0,0,5,0,0)
with 100 vertices and 576 arcs.

Chapter-number grid (top-left is (1,1), top row is the attic):

59	83	15	10	57	48	7	52	45	54
97	11	58	82	16	9	46	55	6	51
84	60	96	14	47	56	49	8	53	44
12	98	81	86	95	17	28	43	50	5
61	85	13	18	27	79	94	4	41	30
99	70	26	80	87	1	42	29	93	3
25	62	88	69	19	36	78	2	31	40
71	65	20	23	89	68	34	37	77	92
63	24	66	73	35	22	90	75	39	32
...	72	64	21	67	74	38	33	91	76

chapters: 99, repeated cells: 0
unvisited cell: (1,10) <- clinamen
non-knight move: ch.65 (2,8) -> ch.66 (3,9), offset (1,1)

14. A genuine complete tour. Perec's path is a knight's tour with a deliberate scar: ninety-nine cells, one illegal diagonal step, one room left forever empty. It is fair to ask whether the board forced his hand—whether a 10×10 board admits any flawless tour of all one hundred cells at all. It does, and to see it we now set the novel aside and let the program find such a tour on its own, on the very same board.

The oldest and simplest rule for the purpose is *Warnsdorff's* (1823): from the cell you stand on, always step to the unvisited cell that has the fewest unvisited neighbors of its own. The idea is to visit the awkward, hard-to-reach cells early, while reaching them is still easy, and to save the roomy ones for last. On the 10×10 board the rule walks clean through all hundred cells without ever having to turn back; so, fittingly, we start it at Perec's own central landing (6,6) and let it finish what he chose to leave undone.

15. The rule needs, for each cell, the list of its knight-neighbors, which we read straight off the board's arcs. Vertex names are coordinates "row.col", so a small reverse map *rev* turns a name back into a **cell**, and *nbr*[*c*] collects the cells a knight can reach from *c*.

⟨Generate and print a complete tour 15⟩ ≡

```
fmt.Fprint(out, "\nPEREC: a genuine complete knight's tour (Warnsdorff)\n\n")
⟨Build the neighbor lists from the board 16⟩
⟨Grow a tour by Warnsdorff's rule 17⟩
⟨Verify and print the complete tour 19⟩
```

This code is used in section 2.

16. ⟨Build the neighbor lists from the board 16⟩ ≡

```
rev := make(map[string]cell)
for y := 1; y ≤ 10; y++ {
  for x := 1; x ≤ 10; x++ {
    rev[name(cell{x, y})] = cell{x, y}
  }
}
nbr := make(map[cell][]cell)
for v := range g.AllVertices() {
  for a := range v.AllArcs() {
    nbr[rev[v.Name]] = append(nbr[rev[v.Name]], rev[a.Tip.Name])
  }
}
```

This code is used in section 15.

17. We start at (6,6), mark it walked, and grow the path one cell at a time. At each step we scan the unvisited neighbors of the current cell and keep the one with the *fewest* onward moves. Should the scan ever find nothing, Warnsdorff has run into a dead end and we stop; but on this board it never does.

⟨ Grow a tour by Warnsdorff's rule 17 ⟩ ≡

```

start := cell{6,6}
walked := map[cell]bool{start:true}
full := []cell{start}
for len(full) < 100 {
  cur := full[len(full) - 1]
  next, fewest, found := cell{ }, 99, false
  for _, n := range nbr[cur] {
    if walked[n] {
      continue
    }
    ⟨ Let onward count n's unvisited neighbors 18 ⟩
    if onward < fewest {
      fewest, next, found = onward, n, true
    }
  }
  if ¬found {
    break
  }
  walked[next] = true
  full = append(full, next)
}

```

This code is used in section 15.

18. This is the crux of the rule: how crowded the cell n still is, measured as the number of its neighbors not yet walked.

⟨ Let onward count n 's unvisited neighbors 18 ⟩ ≡

```

onward := 0
for _, m := range nbr[n] {
  if ¬walked[m] {
    onward++
  }
}

```

This code is used in section 17.

19. We verify the finished tour on the board's own arcs, exactly as we did Perec's, and this time expect no flaw at all: a hundred cells, every step a knight's move, and—unlike the novel—no clinamen. The step-number grid is laid out like the facade, so it can be read against the chapter grid above.

⟨ Verify and print the complete tour 19 ⟩ ≡

```

grid := make(map[cell]int)
var flaws int
for k, c := range full {
    grid[c] = k + 1
    if k > 0 ∧ ¬adj[[2]string{name(full[k - 1]), name(c)}] {
        flaws++
    }
}
⟨ Print the complete-tour grid 20 ⟩
fmt.Fprintf(out, "cells visited: %d, non-knight moves: %d\n", len(full), flaws)
if len(full) == 100 ∧ flaws == 0 {
    fmt.Fprintf(out, "=> a flawless knight's tour of all 100 cells (no clinamen).\n")
}

```

This code is used in section 15.

20. ⟨ Print the complete-tour grid 20 ⟩ ≡

```

fmt.Fprintf(out, "Step-number grid (top-left is (1,1), top row is the attic):\n\n")
for y := 1; y ≤ 10; y++ {
    fmt.Fprintf(out, "    ")
    for x := 1; x ≤ 10; x++ {
        fmt.Fprintf(out, "%3d", grid[cell{x, y}])
    }
    fmt.Fprintf(out, "\n")
}
fmt.Fprintf(out, "\n")

```

This code is used in section 19.

21. Run it and the program prints this for the complete tour: the step-number grid, then a line confirming that all hundred cells were visited with not one non-knight move. The 1 sits at the central landing (6,6), right where Perec began; from there the rule reaches every room, the bottom-left cellar included.

PEREC: a genuine complete knight's tour (Warnsdorff)

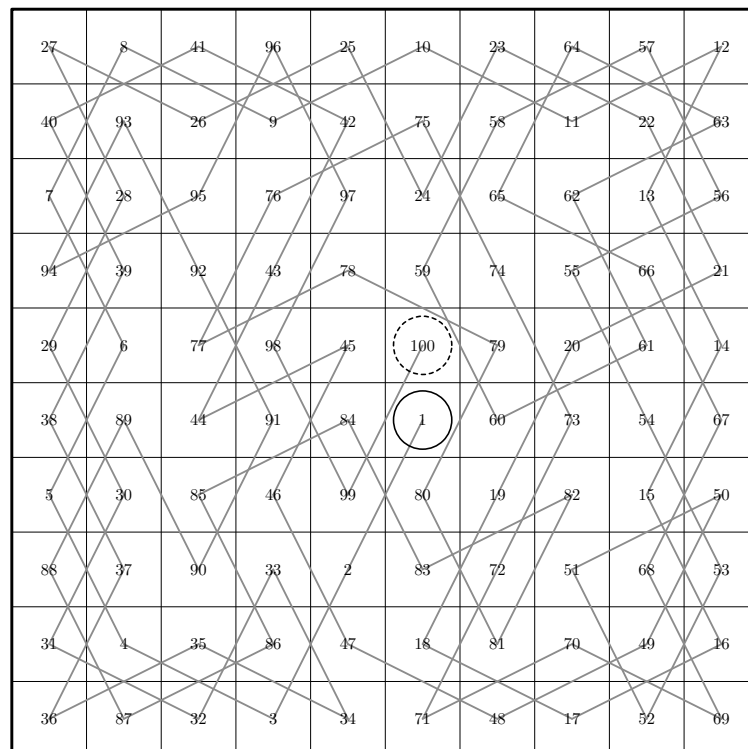
Step-number grid (top-left is (1,1), top row is the attic):

27	8	41	96	25	10	23	64	57	12
40	93	26	9	42	75	58	11	22	63
7	28	95	76	97	24	65	62	13	56
94	39	92	43	78	59	74	55	66	21
29	6	77	98	45	100	79	20	61	14
38	89	44	91	84	1	60	73	54	67
5	30	85	46	99	80	19	82	15	50
88	37	90	33	2	83	72	51	68	53
31	4	35	86	47	18	81	70	49	16
36	87	32	3	34	71	48	17	52	69

cells visited: 100, non-knight moves: 0

=> a flawless knight's tour of all 100 cells (no clinamen).

22. And here is that tour drawn out. This time the grid is plain, not a doll's house cut away for a novel: this tour belongs to no story, only to the board. The hundred cells are joined in the order Warnsdorff's rule visited them; the start (6,6), Perec's central landing, is ringed with a solid circle, and the finish (6,5) with a dashed one. Every link is a true knight's move, and no cell is left out—the flawless tour of which Perec's is the deliberate scarring.



23. A Graeco-Latin square. If the knight's tour decides *where to write*, *what to write* is decided by Perec's second constraint. As we said, he gathered 42 lists of material into twenty-one pairs and, by a 10×10 Graeco-Latin square, assigned to each chapter its combination of material. But why order 10? Behind that number lies two centuries of mathematical drama, and Perec chose this square knowing the story.

A Graeco-Latin square is two Latin squares laid one over the other, each cell holding a pair of symbols, so that all the pairs are distinct. The story begins in 1782 with Euler's *problem of the 36 officers*. Can thirty-six officers, one of each of six ranks and six regiments, be drawn up in a 6×6 array so that every row and every column shows each rank once and each regiment once? That is precisely an order-6 Graeco-Latin square. Euler could not build one however he tried, and finally conjectured that no such square exists whenever the order is of the form $4k + 2$ (that is, 2, 6, 10, 14, ...).

Euler's conjecture was half right and half wrong. In 1900 Gaston Tarry proved, by counting every case by hand, that the order-6 square really is impossible—so for 6 Euler was right. But not beyond it. At the April 1959 meeting of the American Mathematical Society in New York, Bose, Shrikhande and Parker announced that for every order of the form $4k + 2$ except 2 and 6—that is, 10, 14, 18, ...—a Graeco-Latin square exists. Parker found an order-10 square in about an hour's search on a UNIVAC 1206 military computer, one of the earliest combinatorial problems solved on a digital computer. The three were nicknamed *Euler's spoilers*, and that November the cover of *Scientific American* carried their order-10 square in full colour.

So order 10 is no ordinary number. The very order Euler declared impossible, overturned only two centuries later—the square whose existence had just been proved—is the one Perec took for the skeleton of his novel. It is a choice worthy of a member of Oulipo, who prized the beauty of constraint above all. Now it is time to build the square in earnest and, as Perec did, assign the material to each chapter.

24. For an order that is odd or a prime power, a square is built by a simple formula. For a prime p , say, $L1(i, j) = (i + j) \bmod p$ and $L2(i, j) = (i + 2j) \bmod p$ are already two orthogonal Latin squares. But 10 is even and the formula breaks—which is just what fooled Euler and set Parker to his computer. So we do exactly as Parker did: we run a search that builds a random Latin square and hunts for its orthogonal mate, obtain one square (a matter of seconds on today's machines), and set the result down here as *square*. Each cell is a two-component $[2]\text{int}\{a, b\}$.

(Types and data 3) +=

```
var square = [10][10][2]int {
    {{4, 0}, {2, 1}, {5, 2}, {1, 3}, {3, 4}, {6, 5}, {9, 6}, {0, 7}, {7, 8}, {8, 9}},
    {{3, 3}, {7, 9}, {8, 8}, {0, 4}, {6, 6}, {9, 1}, {2, 5}, {1, 2}, {4, 7}, {5, 0}},
    {{0, 1}, {9, 5}, {6, 3}, {7, 0}, {4, 8}, {1, 9}, {3, 2}, {5, 4}, {8, 6}, {2, 7}},
    {{7, 6}, {3, 7}, {9, 4}, {8, 2}, {0, 9}, {2, 8}, {1, 1}, {6, 0}, {5, 3}, {4, 5}},
    {{2, 4}, {1, 8}, {7, 1}, {6, 7}, {8, 5}, {3, 0}, {5, 9}, {4, 6}, {0, 2}, {9, 3}},
    {{5, 8}, {8, 0}, {0, 5}, {3, 9}, {9, 7}, {7, 2}, {4, 4}, {2, 3}, {6, 1}, {1, 6}},
    {{8, 7}, {4, 2}, {2, 0}, {5, 5}, {7, 3}, {0, 6}, {6, 8}, {9, 9}, {1, 4}, {3, 1}},
    {{1, 5}, {0, 3}, {4, 9}, {2, 6}, {5, 1}, {8, 4}, {7, 7}, {3, 8}, {9, 0}, {6, 2}},
    {{9, 2}, {6, 4}, {3, 6}, {4, 1}, {1, 0}, {5, 7}, {8, 3}, {7, 5}, {2, 9}, {0, 8}},
    {{6, 9}, {5, 6}, {1, 7}, {9, 8}, {2, 2}, {4, 3}, {0, 0}, {8, 1}, {3, 5}, {7, 4}},
}
```

25. Here it is in colour. Like that famous *Scientific American* cover, each cell is split on its diagonal, the upper triangle coloured by the first component and the lower by the second. Because each component is Latin, the ten colours each appear once in every row and column; because the two are orthogonal, no upper-lower pair of colours repeats across the hundred cells. That is the visible proof that this picture is a genuine Graeco-Latin square.

0 4	1 2	2 5	3 1	4 3	5 6	6 9	7 0	8 7	9 8
3 3	9 7	8 8	4 0	6 6	1 9	5 2	2 1	7 4	0 5
0 0	9 9	6 6	7 7	4 4	1 1	3 3	5 5	8 8	2 2
6 7	7 3	4 9	2 8	9 0	8 2	1 1	0 6	3 5	5 4
2 4	8 1	1 7	7 6	5 8	0 3	9 5	6 4	2 0	3 9
5 8	0 8	5 0	3 3	9 9	7 7	2 2	4 4	3 3	1 1
8 7	4 4	2 2	5 5	7 7	0 0	6 6	9 9	1 1	3 3
1 1	0 0	4 4	2 2	5 5	8 8	7 7	3 3	9 9	6 6
9 9	6 6	3 3	1 1	4 4	0 0	7 7	5 5	2 2	8 8
6 6	5 5	1 1	9 9	2 2	4 4	0 0	8 8	3 3	7 7

26. We do not take on trust that the square we set down is really Graeco-Latin; we check, in the same spirit as for the knight's tour. Three things: is the first component Latin, is the second Latin, and are all 100 pairs distinct (which is exactly the orthogonality of the two components). If all three hold, we hold in our hands the very thing Euler said could not exist.

⟨ Verify the square and print the assignment 26 ⟩ ≡

```
fmt.Fprint(out, "\nPEREC: an_order-10_Graeco-Latin_square\n\n")
⟨ Check that the square is Graeco-Latin 27 ⟩
⟨ Print the square as a grid 28 ⟩
⟨ Assign one couple of lists to chapters 29 ⟩
```

This code is used in section 2.

27. For a component to be Latin means that in each of the ten rows and columns the symbols 0 through 9 each appear once, so it is enough to OR ten bits and see whether they make 1023. Orthogonality we check by putting the 100 pairs into a set and seeing that its size is 100.

⟨ Check that the square is Graeco-Latin 27 ⟩ $\equiv$

```

latinA, latinB := true, true
for i := 0; i < 10; i++ {
  var rA, cA, rB, cB int
  for j := 0; j < 10; j++ {
    rA |= 1 << square[i][j][0]
    cA |= 1 << square[j][i][0]
    rB |= 1 << square[i][j][1]
    cB |= 1 << square[j][i][1]
  }
  if rA ≠ 1023 ∨ cA ≠ 1023 {
    latinA = false
  }
  if rB ≠ 1023 ∨ cB ≠ 1023 {
    latinB = false
  }
}
seen := make(map[[2]int]bool)
for i := 0; i < 10; i++ {
  for j := 0; j < 10; j++ {
    seen[square[i][j]] = true
  }
}
fmt.Fprintf(out, "Are both components Latin squares? A: %v, B: %v\n", latinA, latinB)
fmt.Fprintf(out, "Are all 100 pairs distinct (orthogonal)? %v (%d distinct pairs)\n",
  len(seen) == 100, len(seen))
if latinA ∧ latinB ∧ len(seen) == 100 {
  fmt.Fprint(out, "=> here is the order-10 square Euler said could not exist.\n\n")
}

```

This code is used in section 26.

28. To show the square itself, we print each cell as two digits *ab*. That these hundred pairs run over 00 through 99 exactly once each is guaranteed by the check above.

⟨ Print the square as a grid 28 ⟩ $\equiv$

```

fmt.Fprint(out, "The square (each cell is two components ab): \n\n")
for i := 0; i < 10; i++ {
  fmt.Fprint(out, "\n\n")
  for j := 0; j < 10; j++ {
    fmt.Fprintf(out, "%d%d", square[i][j][0], square[i][j][1])
  }
  fmt.Fprint(out, "\n")
}
fmt.Fprint(out, "\n")

```

This code is used in section 26.

29. Now the two constraints meet. Each cell (x, y) of the building is given a pair (a, b) by the square. Perec bundled his lists of material ten at a time into a couple, and put the a th item of the one list and the b th of the other into the chapter of that room. Because the square is orthogonal, the hundred cells show a hundred combinations exactly once each—any pairing of the two lists meets in the novel exactly once.

Perec kept twenty-one such couples, but his lists in full and their cell-by-cell assignment fill the vast material of his working notebook (the *cahier des charges*). Here, only to show the structure, we use one illustrative couple: ten animals and ten colours.

⟨ Assign one couple of lists to chapters 29 ⟩ $\equiv$

```
animals := [10]string{"cat", "dog", "horse", "fox", "bear", "deer", "rabbit", "wolf", "hawk",
    "mouse"}
colours := [10]string{"red", "orange", "yellow", "green", "blue", "indigo", "violet", "black",
    "white", "gray"}
fmt.Fprint(out, "One couple (animal, colour) assigned to chapters (first eight):\n\n")
used := make(map[[2]int]bool)
for k, c := range tour {
    p := square[c.y - 1][c.x - 1]
    used[p] = true
    if k < 8 {
        fmt.Fprintf(out, "ch.%2d(%d,%d): %s, %s\n",
            k + 1, c.x, c.y, animals[p[0]], colours[p[1]])
    }
}
```

⟨ Show the missing combination is the clinamen's 30 ⟩

This code is used in section 26.

30. The ninety-nine chapters use ninety-nine distinct combinations. Exactly one of the hundred is missing—the combination the unvisited clinamen cell (1, 10) would have held. The blemish in the tour that fixes the order has taken away one of the things to write as well.

⟨ Show the missing combination is the clinamen's 30 ⟩ $\equiv$

```
fmt.Fprintf(out, "\nCombinations used: %d (of 100)\n", len(used))
q := square[9][0] // square[10 - 1][1 - 1], clinamen cell (1, 10)
fmt.Fprintf(out, "Missing combination: %s, %s < - the share of clinamen cell (1, 10)\n",
    animals[q[0]], colours[q[1]])
```

This code is used in section 29.

31. The output for the square continues like this: the verification result, the square itself, and one illustrative couple assigned to chapters. Ninety-nine combinations are used, and the last two lines confirm that the one share of the clinamen cell is missing.

PEREC: an order-10 Graeco-Latin square

```
Are both components Latin squares?  A: true, B: true
Are all 100 pairs distinct (orthogonal)?  true (100 distinct pairs)
=> here is the order-10 square Euler said could not exist.
```

The square (each cell is two components ab):

```
40 21 52 13 34 65 96 07 78 89
33 79 88 04 66 91 25 12 47 50
01 95 63 70 48 19 32 54 86 27
76 37 94 82 09 28 11 60 53 45
24 18 71 67 85 30 59 46 02 93
58 80 05 39 97 72 44 23 61 16
87 42 20 55 73 06 68 99 14 31
15 03 49 26 51 84 77 38 90 62
92 64 36 41 10 57 83 75 29 08
69 56 17 98 22 43 00 81 35 74
```

One couple (animal, colour) assigned to chapters (first eight):

```
ch. 1 (6,6): wolf, yellow
ch. 2 (8,7): mouse, gray
ch. 3 (10,6): dog, violet
ch. 4 (8,5): bear, violet
ch. 5 (10,4): bear, indigo
ch. 6 (9,2): bear, black
ch. 7 (7,1): mouse, violet
ch. 8 (8,3): deer, blue
```

combinations used: 99 (of 100)

missing combination: rabbit, gray <- the share of clinamen cell (1,10)

32. Index.

a: [5](#), [16](#), [24](#).
ab: [28](#).
abs: [11](#), [12](#).
adj: [5](#), [8](#), [19](#).
AllArcs: [5](#), [16](#).
AllVertices: [5](#), [16](#).
animals: [29](#), [30](#).
append: [8](#), [9](#), [16](#), [17](#).
b: [11](#), [24](#).
Board: [2](#), [4](#), [5](#), [9](#).
bool: [5](#), [17](#), [27](#), [29](#).
breaks: [8](#), [11](#).
bufio: [2](#).
c: [6](#), [8](#), [15](#), [19](#), [29](#).
cA: [27](#).
cB: [27](#).
cell: [3](#), [6](#), [9](#), [10](#), [15](#), [16](#), [17](#), [19](#), [20](#).
ch: [10](#).
chapterAt: [8](#), [9](#), [10](#).
colours: [29](#), [30](#).
cur: [17](#).
dx: [11](#).
dy: [11](#).
err: [4](#).
Fatalf: [4](#).
fewest: [17](#).
flaws: [19](#).
Flush: [2](#).
fmt: [6](#), [7](#), [10](#), [11](#), [15](#), [19](#), [20](#), [26](#), [27](#), [28](#), [29](#), [30](#).
found: [17](#).
Fprint: [7](#), [10](#), [15](#), [19](#), [20](#), [26](#), [27](#), [28](#), [29](#).
Fprintf: [7](#), [10](#), [11](#), [19](#), [20](#), [27](#), [28](#), [29](#), [30](#).
full: [17](#), [19](#).
g: [4](#), [5](#), [7](#), [16](#).
gbbasic: [4](#).
grid: [19](#), [20](#).
i: [24](#), [27](#), [28](#).
ID: [7](#).
int: [3](#), [8](#), [12](#), [19](#), [24](#), [27](#), [29](#).
j: [24](#), [27](#), [28](#).
k: [3](#), [8](#), [19](#), [29](#).
L1: [24](#).
L2: [24](#).
latinA: [27](#).
latinB: [27](#).
len: [11](#), [17](#), [19](#), [27](#), [30](#).
log: [4](#).
M: [7](#).
m: [11](#), [18](#).
main: [2](#).
make: [5](#), [8](#), [16](#), [19](#), [27](#), [29](#).
missing: [9](#), [11](#).
mod: [24](#).
N: [7](#).
n: [12](#), [17](#), [18](#).
Name: [5](#), [16](#).
name: [6](#), [8](#), [9](#), [10](#), [16](#), [19](#).
nbr: [15](#), [16](#), [17](#), [18](#).
NewWriter: [2](#).
next: [17](#).
onward: [17](#), [18](#).
os: [2](#).
out: [2](#), [7](#), [10](#), [11](#), [15](#), [19](#), [20](#), [26](#), [27](#), [28](#), [29](#), [30](#).
p: [11](#), [24](#), [29](#).
piece: [4](#).
q: [11](#), [30](#).
rA: [27](#).
rB: [27](#).
repeats: [8](#), [11](#).
rev: [15](#), [16](#).
seen: [8](#), [9](#), [10](#), [27](#).
Sprintf: [6](#).
sqs: [2](#).
square: [24](#), [27](#), [28](#), [29](#), [30](#).
start: [17](#).
Stdout: [2](#).
string: [5](#), [6](#), [8](#), [16](#), [19](#), [29](#).
Tip: [5](#), [16](#).
tour: [3](#), [8](#), [11](#), [29](#).
used: [29](#), [30](#).
v: [5](#), [16](#).
walked: [17](#), [18](#).
x: [3](#), [5](#), [6](#), [9](#), [10](#), [11](#), [16](#), [20](#), [29](#).
y: [3](#), [5](#), [6](#), [9](#), [10](#), [11](#), [16](#), [20](#), [29](#).

- ⟨Assign one couple of lists to chapters 29⟩ Used in section 26
- ⟨Build the knight board 4⟩ Used in section 2
- ⟨Build the neighbor lists from the board 16⟩ Used in section 15
- ⟨Check that the square is Graeco-Latin 27⟩ Used in section 26
- ⟨Extract the board's adjacency 5⟩ Used in section 2
- ⟨Find non-adjacent links and repeated cells 8⟩ Used in section 7
- ⟨Find the unvisited cell 9⟩ Used in section 7
- ⟨Generate and print a complete tour 15⟩ Used in section 2
- ⟨Grow a tour by Warnsdorff's rule 17⟩ Used in section 15
- ⟨Let *onward* count n 's unvisited neighbors 18⟩ Used in section 17
- ⟨Print the chapter-number grid 10⟩ Used in section 7
- ⟨Print the complete-tour grid 20⟩ Used in section 19
- ⟨Print the square as a grid 28⟩ Used in section 26
- ⟨Print the verification result 11⟩ Used in section 7
- ⟨Show the missing combination is the clinamen's 30⟩ Used in section 29
- ⟨Types and data 3⟩ Used in section 2
- ⟨Verify and print the complete tour 19⟩ Used in section 15
- ⟨Verify and print the tour 7⟩ Used in section 2
- ⟨Verify the square and print the assignment 26⟩ Used in section 2

PEREC

	Section	Page
The knight's tour	1	1
A genuine complete tour	14	6
A Graeco-Latin square	23	10
Index	32	15